

Exercise Sheet 2: AOA and Discrete-Time Model

The notations are the same as in the course.

Exercise 1. Let $(\xi_n)_{n \in \mathbb{N}^*}$ be a sequence of independent and identically distributed random variables (i.i.d) with zero mean and finite variance. We consider the natural filtration $\mathbb{F}$ given by $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and, $\forall n \in \mathbb{N}^*$,

$$\mathcal{F}_n = \sigma(\xi_k, 1 \leq k \leq n).$$

We define the process $X = (X_n)_{n \in \mathbb{N}}$ by

$$X_0 = 0, \quad \text{and} \quad X_n = \sum_{k=1}^n \xi_k \quad \forall n \in \mathbb{N}^*.$$

1. Show that X is an $\mathbb{F}$ -martingale.

2. Show that $\forall n \in \mathbb{N}$,

$$\mathbb{E}[X_{n+1}^2 | \mathcal{F}_n] = X_n^2 + \mathbb{E}[\xi_{n+1}^2].$$

Deduce that there exists an increasing sequence of real numbers $(a_n)_{n \in \mathbb{N}}$ such that $a_0 = 0$ and $(X_n^2 - a_n)_{n \in \mathbb{N}}$ is an $\mathbb{F}$ -martingale.

3. Let $(\eta_n)_{n \in \mathbb{N}^*}$ be another sequence of independent and identically distributed random variables (i.i.d) with zero mean and finite variance such that ξ_k is independent of $\eta_\ell \forall k, \ell \in \mathbb{N}^*$. We introduce the filtration $\mathbb{G}$ given by $\mathcal{G}_0 = \{\emptyset, \Omega\}$ and, $\forall n \in \mathbb{N}^*$,

$$\mathcal{G}_n = \sigma(\xi_k, \eta_\ell, 1 \leq k, \ell \leq n).$$

We define the process $Y = (Y_n)_{n \in \mathbb{N}}$ by

$$Y_0 = 0, \quad \text{and} \quad Y_n = \sum_{\ell=1}^n \eta_\ell \quad \forall n \in \mathbb{N}^*.$$

Show that there exists a sequence of real numbers $(d_n)_{n \in \mathbb{N}}$ such that $d_0 = 0$ and $(X_n Y_n - d_n)_{n \in \mathbb{N}}$ is a $\mathbb{G}$ -martingale.

Exercise 2. Let $\mathbb{F} = (\mathcal{F}_n)_{n \in \mathbb{N}}$ be a filtration on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $(M_n)_{n \in \mathbb{N}}$ be an $\mathbb{F}$ -martingale, and $(\Delta_n)_{n \in \mathbb{N}}$ an adapted and bounded process, i.e., there exists $c > 0$ such that $\forall n \in \mathbb{N}$, $|\Delta_n| \leq c$ almost surely under $\mathbb{P}$. We consider the process $X = (X_n)_{n \in \mathbb{N}}$ defined by

$$X_0 = 0, \quad \text{and} \quad X_n = \sum_{k=1}^n \Delta_{k-1} (M_k - M_{k-1}) \quad \forall n \in \mathbb{N}^*.$$

Show that X is an $\mathbb{F}$ -martingale.

Exercise 3. Three following definitions of an arbitrage opportunity are given:

- (i) We say that a market has an arbitrage opportunity if there exists a self-financing portfolio V such that

$$V_0 = 0, \quad \mathbb{P}(V_T \geq 0) = 1, \quad \text{and} \quad \mathbb{P}(V_T > 0) > 0;$$

- (ii) We say that a market has an arbitrage opportunity if there exists a self-financing portfolio V such that

$$V_0 \leq 0, \quad \mathbb{P}(V_T \geq 0) = 1, \quad \text{and} \quad \mathbb{P}(V_T > 0) > 0;$$

- (iii) We say that a market has an arbitrage opportunity if there exists a self-financing portfolio V such that

$$V_0 \leq 0, \quad \mathbb{P}(V_T \geq 0) = 1, \quad \text{and} \quad (V_0 \neq 0 \text{ or } \mathbb{P}(V_T \neq 0) > 0).$$

Prove that these three definitions are equivalent. (*Definition (i) is the one seen in class.*)

Exercise 4. Show that a strategy θ is self-financing if and only if

$$\langle \theta_{t_{k-1}}, \tilde{S}_{t_{k-1}} \rangle = \langle \theta_{t_k}, \tilde{S}_{t_{k-1}} \rangle$$

$$\forall k \in \{2, \dots, N\}.$$

Exercise 5. Consider a market with $d = 1$, and let an investor always allocate half of their portfolio value to the risky asset and the other half to the risk-free asset.

1. Write the recurrence formula for θ_{t_k} in terms of $\theta_{t_{k-1}}$ and $S_{t_{k-1}}$.
2. Write the recurrence formula for V_{t_k} in terms of $V_{t_{k-1}}$, $S_{t_{k-1}}$, and S_{t_k} .

Exercise 6. Consider a market with $d = 1$, $r = \ln(1.06)$, $N = 2$, $t_0 = 0$, $t_1 = 1$, and $t_2 = 2$, such that $S_{t_0}^0 = 100$, $S_{t_0}^1 = 100$, $S_{t_1}^1 = 120$, and $S_{t_2}^1 = 100$. Let an investor have a self-financing portfolio with value V such that $V_{t_0} = 1000$, $V_{t_1} = 1060$, and $V_{t_2} = 1042$.

1. Calculate $S_{t_k}^0$ and $\tilde{V}_{t_k}$ for $k = 0, 1, 2$.
2. Calculate θ_{t_k} for $k = 1, 2$.
3. Compare the investor's strategy with a strategy involving only the risk-free asset.

Exercise 7. Consider a forward contract with maturity $T > 0$ on an asset S paying a single dividend of amount $D > 0$ at time $\bar{t} \in (0, T)$ (D and $\bar{t}$ are deterministic). Give the value F of the forward. Generalize to the case of an asset S paying m dividends of amounts $D_1, \dots, D_m > 0$ at respective times $0 < \bar{t}_1 < \dots < \bar{t}_m < T$.

Exercise 8. For $S, K > 0$, show that $\forall t \in [0, T]$:

1. $(S - Ke^{-r(T-t)})^+ \leq C_t(S, T, K) \leq S$;

$$2. (S - Ke^{-r(T-t)})^- \leq P_t(S, T, K) \leq Ke^{-r(T-t)}.$$

Exercise 9. For $S > 0$, show that $\forall t \in [0, T]$:

1. The function $K > 0 \mapsto C_t(S, T, K)$ is convex;
2. The function $K > 0 \mapsto P_t(S, T, K)$ is convex.

Exercise 10. Consider a derivative product on an asset S with payoff

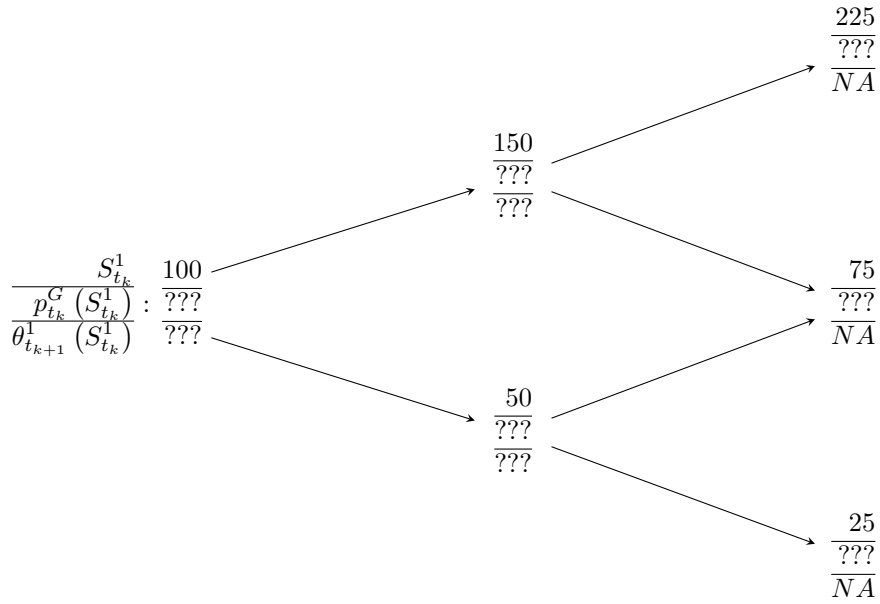
$$\min(\max(S_T, K_1), K_2), \quad K_1 < K_2.$$

1. Plot the payoff of this instrument at maturity T .
2. Express the price at time 0 in terms of the prices of two well-chosen calls.

Exercise 11. Plot the payoff of the following strategies at maturity.

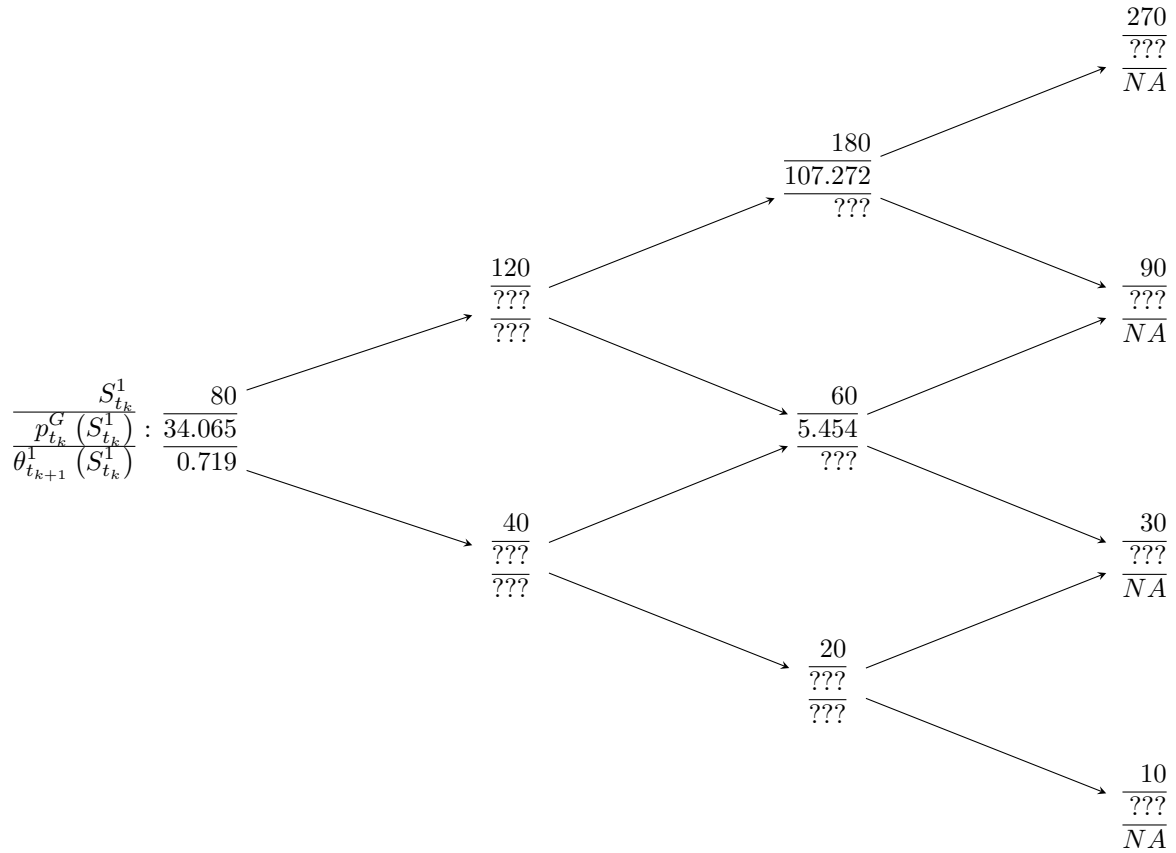
1. Sale of an asset, purchase of two calls with strike K (straddle).
2. Purchase of a call and a put with the same strike K (this strategy is also called a "straddle": why?).
3. Purchase of a call and two puts with strike K (strip).
4. Purchase of two calls and a put with strike K (strap).
5. Purchase of a call with strike K_1 and a put with strike K_2 . Distinguish the three cases $K_1 < K_2$, $K_1 = K_2$, $K_1 > K_2$ (strangle).
6. Purchase of a call with strike K_1 and sale of a call with strike $K_2 > K_1$ (bull spread).
7. Purchase of a call with strike K_2 and sale of a call with strike $K_1 < K_2$ (bear spread).
8. Purchase of a call with strike K_1 , sale of two calls with strike K_2 , purchase of a call with strike K_3 , where $K_1 < K_2 < K_3$ (butterfly spread).

Exercise 12. Consider a binomial tree with 2 steps with $t_0 = 0$, $N = 2$, $T = 4$, $S_{t_0}^1 = 100$, $u = 1.5$, $d = 0.5$, and $r = -\frac{1}{2} \ln(0.8)$. This tree is used for evaluating a European call with maturity T and strike $K = 100$. Calculate the risk-neutral probability q and fill in the missing parts of the following tree:



Exercise 13. Consider a trinomial tree with $t_0 = 0$, $N = T = 3$, $S_{t_0}^1 = 80$, $u = 1.5$, $d = 0.5$, and $r = \ln(1.1)$. This tree is used for evaluating a European call with maturity T and strike $K = 80$.

1. Calculate the risk-neutral probability q and fill in the missing parts of the following tree:



2. Consider for instance the scenario $80 \rightarrow 120 \rightarrow 60 \rightarrow 90$ and show how you can replicate the option. Same question for the scenario $80 \rightarrow 120 \rightarrow 60 \rightarrow 30$.

Exercise 14. Show that for a trinomial tree, the risk-neutral probability is not unique.

Exercise 15. Consider a complete market with a risky asset for $d = 1$, admitting a risk-neutral probability. Suppose that for all $k \in \mathbb{N}$, the variable $S_{t_k}^1$ takes values in a finite set. Compare the discounted prices of a derivative product whose payoff corresponds to the indicator of one of the possible values of S_T^1 (Arrow-Debreu asset) with the risk-neutral probability. Generalize for $d > 1$.